

AP Precalculus

Course Outline • Mathlify Summer Program

Course Overview

This course prepares students for success in AP Precalculus by strengthening their understanding of functions, mathematical modeling, and analytical reasoning. Students develop fluency with polynomial, rational, exponential, logarithmic, and trigonometric functions while practicing AP-style multiple choice and free-response questions. The goal is to ensure students enter the school year confident, prepared, and ready to perform at a high level on the AP exam.

Course Units

Unit 1 — Polynomial & Rational Functions

- Polynomial behavior and end behavior
- Zeros, multiplicity, and graph interpretation
- Rational functions and asymptotes
- Function transformations and modeling

Unit 2 — Exponential & Logarithmic Functions

- Exponential growth and decay
- Logarithmic properties and inverses
- Solving exponential and logarithmic equations
- Real-world modeling

Unit 3 — Trigonometric Functions

- Unit circle and radian measure
- Graphing sine, cosine, tangent
- Transformations and periodic models
- Trigonometric identities and equations